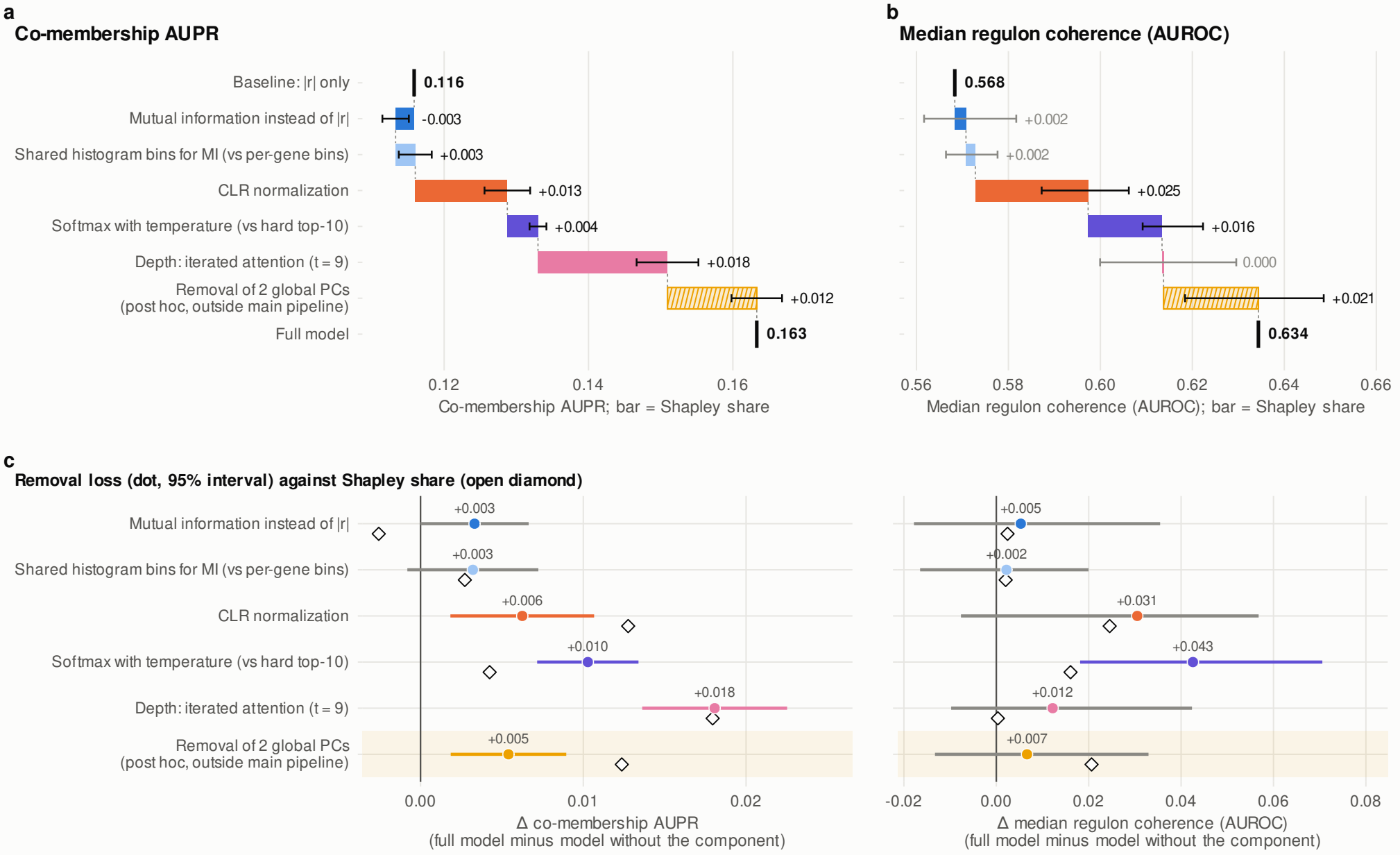
CLR normalization and softmax drive the coherence gain; depth and CLR drive pooled AUPR

M3D, 466 experiments; RegulonDB strong/confirmed evidence. a, b: exact Shapley shares over all 64 on/off combinations of six components, added in pipeline order from the |r| baseline to the full model; whiskers 95% intervals (gene jackknife for AUPR, regulon bootstrap for coherence), grey where they include 0. c: loss when one component is removed from the full model (dot, interval) against its Shapley share (open diamond); a gap is an interaction. Largest interactions: softmax on coherence (removal +0.043 vs Shapley +0.016) and depth on coherence (removal +0.012 vs Shapley 0.000): both matter most when the other components are present.



Hatched: PC removal was chosen after inspecting Supplementary Note 7 and is not part of the main pipeline. 'Off' alternatives: |r|, per-gene Freedman-Diaconis bins, raw dependence, hard top-10 neighbours, no attention, no PC removal.